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compact1, compact2, compact3

java.io

Class PipedReader

java.lang.Object
    java.io.Reader
        java.io.PipedReader

All Implemented Interfaces:

`Closeable`, `AutoCloseable`, `Readable`

```
public class PipedReader
extends Reader
```

Piped character-input streams.

Since:

JDK1.1

Field Summary

Fields inherited from class java.io.Reader

`lock`

Constructor Summary

Constructors

Constructor	Description
`PipedReader()`	

	Creates a PipedReader so that it is not yet **connected**.
PipedReader(int pipeSize)	Creates a PipedReader so that it is not yet **connected** and uses the specified pipe size for the pipe's buffer.
PipedReader(**PipedWriter** src)	Creates a PipedReader so that it is connected to the piped writer src.
PipedReader(**PipedWriter** src, int pipeSize)	Creates a PipedReader so that it is connected to the piped writer src and uses the specified pipe size for the pipe's buffer.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
void	**close**()	Closes this piped stream and releases any system resources associated with the stream.
void	**connect**(**PipedWriter** src)	Causes this piped reader to be connected to the piped writer src.
int	**read**()	Reads the next character of data from this piped stream.
int	**read**(char[] cbuf, int off, int len)	Reads up to len characters of data from this piped stream into an array of characters.
boolean	**ready**()	Tell whether this stream is ready to be read.

Methods inherited from class java.io.Reader

mark, markSupported, read, read, reset, skip

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Detail

PipedReader

```
public PipedReader(PipedWriter src)
            throws IOException
```

Creates a PipedReader so that it is connected to the piped writer src. Data written to src will then be available as input from this stream.

Parameters:

src - the stream to connect to.

Throws:

IOException - if an I/O error occurs.

PipedReader

```
public PipedReader(PipedWriter src,
                   int pipeSize)
            throws IOException
```

Creates a PipedReader so that it is connected to the piped writer src and uses the specified pipe size for the pipe's buffer. Data written to src will then be available as input from this stream.

Parameters:

src - the stream to connect to.

pipeSize - the size of the pipe's buffer.

Throws:

IOException - if an I/O error occurs.

IllegalArgumentException - if pipeSize <= 0.

Since:

1.6

PipedReader

```
public PipedReader()
```

Creates a `PipedReader` so that it is not yet connected. It must be connected to a `PipedWriter` before being used.

PipedReader

```
public PipedReader(int pipeSize)
```

Creates a `PipedReader` so that it is not yet `connected` and uses the specified pipe size for the pipe's buffer. It must be connected to a `PipedWriter` before being used.

Parameters:

`pipeSize - the size of the pipe's buffer.`

Throws:

`IllegalArgumentException - if pipeSize <= 0.`

Since:

`1.6`

Method Detail

connect

```
public void connect(PipedWriter src)
             throws IOException
```

Causes this piped reader to be connected to the piped writer `src`. If this object is already connected to some other piped writer, an `IOException` is thrown.

If `src` is an unconnected piped writer and `snk` is an unconnected piped reader, they may be connected by either the call:

`snk.connect(src)`

or the call:

`src.connect(snk)`

The two calls have the same effect.

Parameters:

src - The piped writer to connect to.

Throws:

IOException - if an I/O error occurs.

read

```
public int read()
         throws IOException
```

Reads the next character of data from this piped stream. If no character is available because the end of the stream has been reached, the value -1 is returned. This method blocks until input data is available, the end of the stream is detected, or an exception is thrown.

Overrides:

read in class Reader

Returns:

the next character of data, or -1 if the end of the stream is reached.

Throws:

IOException - if the pipe is broken, unconnected, closed, or an I/O error occurs.

read

```
public int read(char[] cbuf,
                int off,
                int len)
         throws IOException
```

Reads up to len characters of data from this piped stream into an array of characters. Less than len characters will be read if the end of the data stream is reached or if len exceeds the pipe's buffer size. This method blocks until at least one character of input is available.

Specified by:

read in class Reader

Parameters:

cbuf - the buffer into which the data is read.

off - the start offset of the data.

len - the maximum number of characters read.

Returns:

the total number of characters read into the buffer, or -1 if there is no more data because the end of the stream has been reached.

Throws:

IOException - if the pipe is broken, unconnected, closed, or an I/O error occurs.

ready

```
public boolean ready()
              throws IOException
```

Tell whether this stream is ready to be read. A piped character stream is ready if the circular buffer is not empty.

Overrides:

ready in class Reader

Returns:

True if the next read() is guaranteed not to block for input, false otherwise. Note that returning false does not guarantee that the next read will block.

Throws:

IOException - if the pipe is broken, unconnected, or closed.

close

```
public void close()
           throws IOException
```

Closes this piped stream and releases any system resources associated with the stream.

Specified by:

close in interface Closeable

Specified by:

close in interface AutoCloseable

Specified by:

close in class Reader

Throws:

IOException - if an I/O error occurs.

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
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